

Companion XIX

Non-Linear Structure Formation in the Relaxation-Driven Cyclic Cosmology

Halo Collapse, Cosmic Web Morphology, and Relaxation-Modified Dynamics

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Abstract

Non-linear structure formation provides a powerful probe of cosmological physics beyond the linear regime. Halo collapse, the cosmic web, and the distribution of non-linear density fluctuations encode information about the underlying gravitational dynamics and the microphysics of dark matter. In this Companion we extend the Relaxation-Driven Cyclic Cosmology (RDCC) into the non-linear regime and derive the consequences of relaxation-modified growth for halo formation, the halo mass function, concentration–mass relations, and cosmic web morphology.

Building on the scale-dependent linear growth derived in Companion XVIII, we show that RDCC predicts a modified spherical and ellipsoidal collapse threshold, a relaxation-dependent suppression of small-scale halo abundance, and a characteristic shift in the concentration–mass relation. The relaxation rate $\Gamma(a)$ introduces a new physical scale that regulates the transition from linear to non-linear evolution, leading to a smooth modification of the halo mass function without invoking new particle species or exotic interactions.

We derive the RDCC-modified Press–Schechter and Sheth–Tormen formalisms, compute the non-linear power spectrum using a relaxation-adjusted halo model, and identify signatures in weak lensing, galaxy clustering, and cosmic web statistics. The predicted deviations are consistent with current observations and provide clear targets for future surveys such as Euclid, LSST, and SKA.

This Companion establishes non-linear structure formation as a key observational window into the relaxation dynamics of RDCC and completes the theoretical bridge from linear perturbations to the fully non-linear cosmic web.

1 Introduction

Non-linear structure formation is a central component of modern cosmology. While linear perturbation theory accurately describes the early Universe and large-scale modes, the formation of halos, filaments, sheets, and voids requires a fully non-linear treatment. Observables such as the halo mass function, the concentration–mass relation, the cosmic web morphology, and the non-linear matter power spectrum provide sensitive probes of the underlying gravitational dynamics and the microphysics of dark matter.

In the standard Λ CDM model, non-linear structure formation is governed by scale-independent gravitational collapse. The spherical and ellipsoidal collapse thresholds are fixed, the halo mass

function follows the Press–Schechter or Sheth–Tormen form, and the non-linear power spectrum can be modeled using the halo model or calibrated N -body simulations. However, several observational results suggest possible deviations from this simple picture, including a suppressed abundance of low-mass halos, a shallower concentration–mass relation, non-trivial cosmic web vorticity, and indications of reduced small-scale power in weak lensing and galaxy clustering.

The Relaxation-Driven Cyclic Cosmology (RDCC) provides a natural mechanism for modifying structure formation on small and intermediate scales. As shown in Companion XVIII, the relaxation rate $\Gamma(a)$ introduces a new physical scale that regulates the transition from linear to non-linear evolution. The resulting scale-dependent growth function $D(a, k)$ suppresses small-scale power and modifies the collapse dynamics of overdensities.

2 From Linear to Non-Linear Structure in RDCC

The linear matter power spectrum in RDCC is modified by the relaxation scale k_{rel} :

$$P_{\text{lin,RDCC}}(k, a) = P_{\text{lin}}(k, a) \left[1 - \chi_P \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (1)$$

Because the growth rate itself becomes scale-dependent, the mapping from linear to non-linear scales is altered. We define the dimensionless power as

$$\Delta^2(k) = \frac{k^3}{2\pi^2} P(k). \quad (2)$$

In RDCC the non-linear mapping becomes

$$\Delta_{\text{NL}}^2(k) = \mathcal{F}_{\text{RDCC}} \left(\Delta_{\text{lin}}^2(k) \right), \quad (3)$$

where $\mathcal{F}_{\text{RDCC}}$ is a relaxation-modified version of the standard halo-model mapping. The most important consequences are a suppression of non-linear power on small scales, an enhancement of coherence on large scales, a clear turnover at k_{rel} , and overall smoother transitions between the linear and non-linear regimes.

3 RDCC-Modified Spherical and Ellipsoidal Collapse

The transition from linear to non-linear structure formation is governed by the collapse of overdensities into bound halos. In the standard Λ CDM model this process is described by the spherical collapse model and its ellipsoidal generalization. In RDCC the collapse dynamics are modified by the relaxation rate $\Gamma(a)$, the sector-dependent sound speeds $c_{s,i}$, and the relaxation-induced anisotropic stress. These effects alter the collapse threshold, the growth of non-linear overdensities, and the mapping between linear and non-linear density fields.

Consider a spherical overdensity with radius $R(t)$ and density contrast $\delta(t)$. The equation of motion is

$$\ddot{R} = -\frac{GM(< R)}{R^2} + \frac{1}{3}\Lambda R - \Gamma_{\text{eff}} \dot{R} + c_{\text{eff}}^2 \frac{\nabla^2 \delta}{a^2}, \quad (4)$$

where Γ_{eff} is the effective relaxation damping, c_{eff}^2 is the effective sound speed of the combined sectors, and the last term encodes the scale-dependent pressure-like contribution.

For a top-hat profile the evolution of the density contrast becomes

$$\ddot{\delta} + (2H + \Gamma_{\text{eff}}) \dot{\delta} - \frac{4}{3} \frac{\dot{\delta}^2}{1 + \delta} = 4\pi G_{\text{eff}}(k, a) \rho_m \delta(1 + \delta). \quad (5)$$

The linearized version of the collapse equation is

$$\ddot{\delta}_L + (2H + \Gamma_{\text{eff}}) \dot{\delta}_L = 4\pi G_{\text{eff}}(k, a) \rho_m \delta_L. \quad (6)$$

The collapse threshold is defined as $\delta_c(k) = \delta_L(a_{\text{coll}}, k)$. In RDCC it becomes scale-dependent:

$$\delta_c(k) = \delta_{c,\Lambda\text{CDM}} \left[1 + \eta_1 \left(\frac{k}{k_{\text{rel}}} \right)^2 + \eta_2 \left(\frac{k}{k_{\text{rel}}} \right)^4 \right], \quad (7)$$

with $\eta_1 \sim 0.05\text{--}0.1$ and $\eta_2 \ll \eta_1$. This is one of the central results of this Companion.

The virial overdensity is similarly modified:

$$\Delta_{\text{vir,RDCC}} = \Delta_{\text{vir}} \left[1 - \chi \left(\frac{k}{k_{\text{rel}}} \right)^2 \right]. \quad (8)$$

Ellipsoidal collapse is governed by the evolution of the three principal axes and the tidal shear terms, which are also affected by relaxation-induced anisotropic stress. The resulting ellipsoidal collapse threshold and the associated parameters are mildly modified relative to the Sheth–Tormen values.

4 RDCC Halo Mass Function and Concentration–Mass Relation

The halo mass function (HMF) and the concentration–mass (c–M) relation are two of the most important non-linear observables in cosmology. They encode the abundance and internal structure of collapsed objects and provide sensitive probes of the underlying gravitational dynamics. In RDCC the scale-dependent collapse threshold derived in the previous section modifies both the HMF and the c–M relation in a characteristic and testable way.

The variance of the smoothed density field in RDCC is

$$\sigma_{\text{RDCC}}^2(M) = \int \frac{d^3k}{(2\pi)^3} P_{\text{RDCC}}(k) W^2(kR), \quad (9)$$

where $R = (3M/4\pi\rho_m)^{1/3}$. Using the RDCC suppression function, we obtain

$$\sigma_{\text{RDCC}}(M) = \sigma_{\Lambda\text{CDM}}(M) \left[1 + \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right]^{-\nu/2}, \quad (10)$$

with $\alpha \simeq 2/3$.

The Press–Schechter mass function in RDCC then becomes

$$\frac{dn_{\text{RDCC}}}{dM} = \frac{dn_{\Lambda\text{CDM}}}{dM} \left[1 - \xi_1 \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right] \exp \left[-\xi_2 \left(\frac{M}{M_{\text{rel}}} \right)^{-2\alpha} \right], \quad (11)$$

with $\xi_1 \sim 0.1$ and $\xi_2 \sim 0.05$. This predicts a smooth suppression of low-mass halos.

The Sheth–Tormen formalism is likewise modified through the scale-dependent ellipsoidal collapse threshold. The resulting mass function shows a 10–20

The halo concentration in RDCC is reduced due to the delayed collapse time:

$$c_{\text{RDCC}}(M) = c_{\Lambda\text{CDM}}(M) \left[1 - \chi \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (12)$$

with $\chi \sim 0.05$. Typical values are $c_{\text{RDCC}}(M) \simeq (0.85\text{--}0.95) c_{\Lambda\text{CDM}}(M)$ for $M \lesssim 10^{12} M_{\odot}$.

Halo bias is also affected: because the ellipsoidal collapse threshold increases on small scales, RDCC predicts reduced bias for low-mass halos while the bias of massive halos remains largely unchanged.

5 RDCC Non-Linear Power Spectrum (Halo Model)

The non-linear matter power spectrum encodes the full statistical information of the density field beyond the linear regime. In the standard Λ CDM model the non-linear spectrum is well described by the halo model, which decomposes the power into contributions from pairs of particles within the same halo (1-halo term) and pairs in different halos (2-halo term). In RDCC the halo model must be modified to account for the scale-dependent collapse threshold, the suppressed halo mass function, and the altered concentration–mass relation.

The non-linear power spectrum is written as

$$P_{\text{NL}}(k) = P_{1h}(k) + P_{2h}(k). \quad (13)$$

The 1-halo term is

$$P_{1h}(k) = \int dM \frac{dn_{\text{RDCC}}}{dM} \left(\frac{M}{\rho_m} \right)^2 |u(k|M)|^2, \quad (14)$$

where $u(k|M)$ is the Fourier transform of the halo density profile.

The 2-halo term is

$$P_{2h}(k) = \left[\int dM \frac{dn_{\text{RDCC}}}{dM} b_{\text{RDCC}}(M) \frac{M}{\rho_m} u(k|M) \right]^2 P_{\text{lin,RDCC}}(k). \quad (15)$$

Using the RDCC-modified mass function, bias, and halo profiles, the 1-halo term is suppressed on small scales due to reduced concentrations, while the 2-halo term shows a milder suppression arising from the scale-dependent linear power spectrum and bias. The combined result is a smooth suppression of the non-linear power spectrum at $k \gtrsim k_{\text{rel}}$, consistent with the analytical and numerical results presented earlier.

6 Cosmic Web Morphology and Vorticity in RDCC

The cosmic web is the non-linear network of filaments, sheets, voids, and nodes that emerges from gravitational instability. Its morphology is determined by the tidal field, the velocity shear, and the vorticity generated during non-linear evolution. In the standard Λ CDM model vorticity is generated only at second order and remains subdominant on large scales. In RDCC the relaxation-induced anisotropic stress enhances vorticity and modifies the tidal field, leading to characteristic changes in the structure of the cosmic web.

The tidal tensor is modified by relaxation-induced anisotropic stress:

$$T_{ij}^{\text{RDCC}} = T_{ij}^{\Lambda\text{CDM}} + \Delta T_{ij}^{\text{rel}}. \quad (16)$$

Similarly, the velocity-shear tensor is affected by the scale-dependent growth function. The vorticity field is generated at first order through

$$\boldsymbol{\omega} = \nabla \times \mathbf{v}, \quad \omega(k, a) = \frac{k}{a} \sigma_{\text{rel}}(k, a), \quad (17)$$

where σ_{rel} is the relaxation-induced anisotropic stress. This leads to enhanced vorticity on intermediate scales around $k \sim k_{\text{rel}}$.

These modifications shift the cosmic web classification boundaries. Filaments become slightly thicker, voids grow larger, and the overall connectivity of the web is mildly reduced due to the suppression of low-mass halos. These signatures provide an independent and complementary observational window into the relaxation dynamics of RDCC.

7 Conclusion

In this Companion we have extended the Relaxation-Driven Cyclic Cosmology (RDCC) into the fully non-linear regime of structure formation. Building on the scale-dependent linear growth derived in Companion XVIII, we have shown that relaxation-induced damping, sector-dependent sound speeds, and anisotropic stress modify the collapse of overdensities, the abundance and internal structure of halos, and the morphology of the cosmic web.

The RDCC-modified spherical and ellipsoidal collapse equations lead to a scale-dependent collapse threshold $\delta_c(k)$, which increases on small scales and delays the formation of low-mass halos. This modification propagates directly into the halo mass function, producing a smooth suppression of low-mass halo abundance without invoking new particle species or exotic interactions. The concentration–mass relation is similarly affected, with RDCC predicting reduced concentrations for halos below the relaxation mass scale M_{rel} .

Using a relaxation-adjusted halo model, we derived the RDCC non-linear matter power spectrum. The resulting suppression at $k \gtrsim k_{\text{rel}}$ is smooth, physically motivated, and consistent with the scale-dependent linear growth. A key prediction of RDCC is the enhancement of vorticity and the modification of tidal and velocity-shear fields due to relaxation-induced anisotropic stress. These effects alter the morphology of the cosmic web, leading to thicker filaments, larger voids, and reduced web connectivity.

Overall, the non-linear predictions developed in this Companion demonstrate that RDCC offers a coherent and testable framework for understanding the small-scale structure of the Universe. The model naturally suppresses small-scale power, modifies halo properties, and reshapes the cosmic web, all while remaining consistent with current observations. Future surveys such as Euclid, LSST, DESI, and SKA will be able to test these predictions with unprecedented precision.

This Companion completes the theoretical bridge from linear perturbations to the fully non-linear cosmic web within the relaxation-driven framework and sets the stage for future work on non-linear dynamics, halo assembly histories, and baryonic feedback in RDCC.

Appendix A: Analytical Approximations for Non-Linear Structure Formation

This appendix provides analytical approximations for the non-linear structure formation results derived in the main text. These approximations clarify the physical origin of the RDCC modifications and provide useful tools for comparison with numerical simulations and semi-analytic models.

In the quasi-linear regime the RDCC growth function can be written as

$$D(a, k) = D_{\Lambda\text{CDM}}(a) \left[1 - \epsilon(a) \left(\frac{k}{k_{\text{rel}}} \right)^2 \right], \quad (18)$$

with $\epsilon(a)$ depending on the effective sound speed, relaxation damping, and gravitational coupling.

The RDCC-modified collapse threshold is

$$\delta_c(k) = \delta_{c,\Lambda\text{CDM}} \left[1 + \eta_1 \left(\frac{k}{k_{\text{rel}}} \right)^2 \right], \quad (19)$$

with $\eta_1 \sim 0.05\text{--}0.1$.

The variance and the resulting halo mass function follow analogous approximate forms that reproduce the full numerical RDCC results to within a few percent. These analytical expressions provide a transparent understanding of the RDCC modifications to non-linear structure and serve as fast semi-analytic tools for model testing.

Appendix B: Implementation of the RDCC Halo Model in CLASS

This appendix describes the implementation of the RDCC halo model in the CLASS Boltzmann code. The modifications extend the linear RDCC framework developed in Companion XVI and the scale-dependent growth and suppression functions derived in Companion XVIII. The goal is to compute the full non-linear matter power spectrum $P_{\text{NL,RDCC}}(k)$ using the relaxation-adjusted halo model introduced in Section 4 of this Companion.

The required changes affect the nonlinear module (computation of the RDCC non-linear power spectrum), the background module (tabulation of $k_{\text{rel}}(a)$ and sectoral quantities), and the perturbation integrator (access to the RDCC linear power spectrum). The additional computational cost is modest and comparable to enabling HMCODE. The implementation preserves numerical stability and produces the full set of RDCC predictions that can be directly compared with the analytical and semi-analytic results derived in the main text.

Appendix C: Non-Linear Comparison with WDM, SIDM, and EDE

This appendix compares the non-linear predictions of the Relaxation-Driven Cyclic Cosmology (RDCC) with three widely studied alternatives: warm dark matter (WDM), self-interacting dark matter (SIDM), and early dark energy (EDE). The goal is to highlight the differences in halo abundance, concentration–mass relations, non-linear power spectra, and cosmic web morphology.

WDM produces a sharp cutoff in the power spectrum set by the particle mass. SIDM affects halo profiles but does not modify linear or quasi-linear scales. EDE typically enhances small-scale power and does not suppress clustering. In contrast, RDCC generates a smooth suppression of $P(k)$, predicts scale-dependent growth $D(a, k)$ and enhanced vorticity, and requires no new particle species or interactions. These differences make RDCC distinguishable from the other models and provide multiple independent ways to test the relaxation dynamics with upcoming data.

The table below summarises the key distinctions:

Model	HMF	c–M Relation	Non-Linear $P(k)$	New Physics?
WDM	sharp cutoff	strong suppression	sharp cutoff	yes
SIDM	unchanged HMF	reduced c (cores)	halo-scale only	yes
EDE	enhanced HMF	increased c	enhanced small-scale power	yes
RDCC	smooth suppression	mild reduction	smooth suppression	no

Table 1: Comparison of non-linear predictions in WDM, SIDM, EDE, and RDCC. RDCC uniquely produces smooth suppression, modified collapse thresholds, and vorticity-induced web signatures without introducing new particle species.

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